



Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please
place your student identification label in this box

WA student number: In figures

--	--	--	--	--	--	--	--

In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

--

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	7	
2	7	
3	8	
4	7	
5	7	
6	8	
7	8	
Section 1 Total	52	
Weighted ($\times 0.6731$)	35%	
Section 2 Weighted	65%	
Total	100%	

Section One: Calculator-free

35% (52 Marks)

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(7 marks)

- (a) State the size of the matrix $\begin{bmatrix} 22 & 11 & -11 \end{bmatrix}$.

(1 mark)

- (b) Evaluate

(i) $4 \begin{bmatrix} 6 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} -7 & 11 \\ 5 & 2 \end{bmatrix}$.

(2 marks)

(ii) $\begin{bmatrix} 1 & 7 \\ 4 & 10 \end{bmatrix} \times \begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix}$.

(2 marks)

- (c) Determine the value of p and the value of q given $\begin{bmatrix} p \\ -4 \end{bmatrix} - 3 \begin{bmatrix} 7 \\ q \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$.

(2 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 2**(7 marks)**

(a) Evaluate the following expressions:

(i) $0.25 \times 20 - 8.$

(1 mark)

(ii) $\frac{3}{0.5 \times 8} - \frac{1}{6^2 - 32}.$

(2 marks)

(b) Determine the value of $\sqrt{5b - 3ac - 2}$ when $a = -3$, $b = 6$ and $c = 4$.

(2 marks)

(c) Given that $x^2 - y^2 = (x + y)(x - y)$, evaluate $41^2 - 11^2$.

(2 marks)

Question 3

(8 marks)

In matrix $X = \begin{bmatrix} 5 & 7 & 8 \\ 6 & 7 & 6 \end{bmatrix}$, element $x_{i,j}$ is the number of single people (first row, $i = 1$) or couples (second row, $i = 2$) who attended a dance class on night j .

(a) Show that a total of 17 people attended the dance class on night 1. (1 mark)

(b) Determine $[1 \ 2] \times X$ and explain what information is contained in your answer. (2 marks)

The fee to attend a dance class for one night was \$20 for a single and \$30 for a couple.

(c) Represent the fees in row matrix Y so that the matrix product $Y \times X$ will have a meaningful result. (1 mark)

(d) Determine $Y \times X \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. (3 marks)

(e) Explain what information is contained in the matrix calculated in part (d). (1 mark)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 4**(7 marks)**

- (a) A vertical flagpole, of height 3.6 metres, stands near a vertical lighting tower on a level sports ground. At the instant the length of the shadow of the flagpole is 2.5 metres, the length of shadow of the lighting tower is 7.5 metres. By first sketching the two similar triangles involved in this situation, determine the height of the lighting tower. (3 marks)
- (b) A measurement of 60 millimetres on a construction plan represents a length of 12 metres in a building. Determine the scale factor of the plan in the form $1:n$. (2 marks)
- (c) The straight-line distance between points P and Q on a map drawn to a scale of $1:25\,000$ is 10 centimetres. Determine the actual straight-line distance between the points in kilometres. (2 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 5

(7 marks)

Takeshi lives in a boarding house that charges him \$420 each week for food and lodging.

He has allocated all his weekly income of \$840 to one of five budget categories shown at right.

He always allocates 15% of his weekly income to savings.

	A	B
1	Board	\$420
2	Technology	\$55
3	Car	\$x
4	Savings	\$y
5	Other	\$99
6	Total	\$840

- (a) The boarding house fines residents 5% of their weekly charge for late payment. Determine how much Takeshi would be fined for late payment. **(1 mark)**

- (b) Determine his budget allocation for savings (y). **(1 mark)**

- (c) Takeshi estimates that 40% of his car budget is spent on fuel. Assuming a year of exactly 52 weeks, determine his annual fuel costs. **(3 marks)**

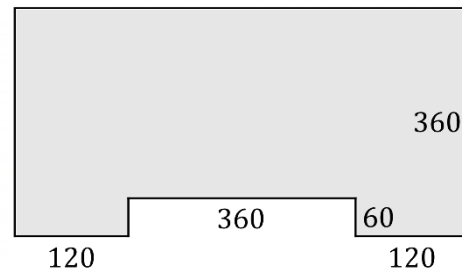
- (d) Takeshi plans to increase the budget allocation for Technology by \$12 and for Other by the same amount. When he does so, which of these two categories will have the largest percentage increase? Justify your answer. **(2 marks)**

Question 6**(8 marks)**

The floor plan of a room, not drawn to scale, is shown. All corners are right-angled and all measurements are in centimetres.

The floor of the room is to be tiled with rectangular tiles that measure 0.3 metres by 0.6 metres.

The tiles are sold in boxes of 10 at a cost of \$90 per box. Only full boxes can be bought.



(a) How long is the longest side of the room, in metres? **(1 mark)**

(b) Determine how much it will cost to buy tiles for the room. **(5 marks)**

Once the floor has been tiled, a thin wooden strip that costs \$5 per metre is to be glued around the edge of the tiled area.

(c) Determine the cost of wooden strip to edge the tiles. **(2 marks)**

Question 7

(8 marks)

Fruit pickers at a farm are paid \$10 per carton of pears plus a wage of \$20 per hour, with both rates increasing by 50% on weekends.

- (a) Determine how much Amana will earn for picking 12 cartons of pears during an 8 hour shift on a Wednesday. (2 marks)
- (b) Determine how much Babbe will earn for picking 8 cartons of pears during a 6 hour shift on a Saturday. (2 marks)
- (c) Carma picked 8 cartons of pears during a 5 hour shift on Sunday and 11 cartons during a second shift on Monday. Given that she earned \$520 for the two days' work, determine how long she worked for on Monday. (4 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Supplementary page

Question number: _____

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Supplementary page

Question number: _____

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

